Urban Planning in West Tripura District Using ArcGIS Pro: A Case Study

Welcome to our presentation on a critical case study in urban planning. We will explore how West Tripura District is navigating the challenges of growth and sustainability using advanced geospatial technology.

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Introduction: Why Urban Planning Matters in West Tripura

West Tripura District, a vibrant region within Tripura, India, faces significant challenges due to rapid urbanization. The delicate balance between development and environmental preservation, especially concerning forests and vegetation, is increasingly critical.

Rapid Urbanization

The district is experiencing accelerated growth, putting immense pressure on its natural landscapes and resources.

Ecological Balance

Maintaining the rich biodiversity and green cover is crucial for the long-term health and livability of the region.

Geospatial Solutions

ArcGIS Pro offers a robust platform for visualizing, analyzing, and understanding complex land cover changes over time, providing invaluable insights for informed decision-making.

Methodology: Harnessing ArcGIS Pro & Satellite Data

Our study employed a rigorous methodology leveraging the advanced capabilities of ArcGIS Pro and satellite imagery to accurately map and analyze land cover changes. We focused on a 35-year period to observe long-term trends.

1

Data Acquisition

Utilized high-resolution Landsat satellite imagery from two distinct periods: 1990 and 2025. These images provided the foundational raw data for our analysis, allowing for a direct comparison of land cover at different points in time.

2

Image Processing

The raw satellite images were pre-processed, including radiometric and atmospheric corrections, to ensure accuracy and comparability between the two timeframes. This step is crucial for minimizing errors in the subsequent analysis.

3

NDBI Classification

Applied NDBI classification in ArcGIS Pro to categorize land cover (Forest, Vegetation, Built-up Areas, Other) using training samples and ground truth. NDBI = (SWIR - NIR) / (SWIR + NIR), where SWIR is Short-Wave Infrared (Band 6) and NIR is Near-Infrared (Band 5), aids built-up area detection.

4

Change Detection Analysis

Performed a robust change detection analysis between the 1990 and 2025 classified maps. This process identified specific areas where land cover transformations occurred, highlighting regions of deforestation, urbanization, or changes in vegetation density.

5

Data Visualization

The analyzed data was presented using various visualization techniques, including four distinct chart types (bar charts, pie charts, line graphs, and heatmaps) to effectively illustrate temporal changes in each land cover category. Spatial changes were depicted through side-by-side maps.

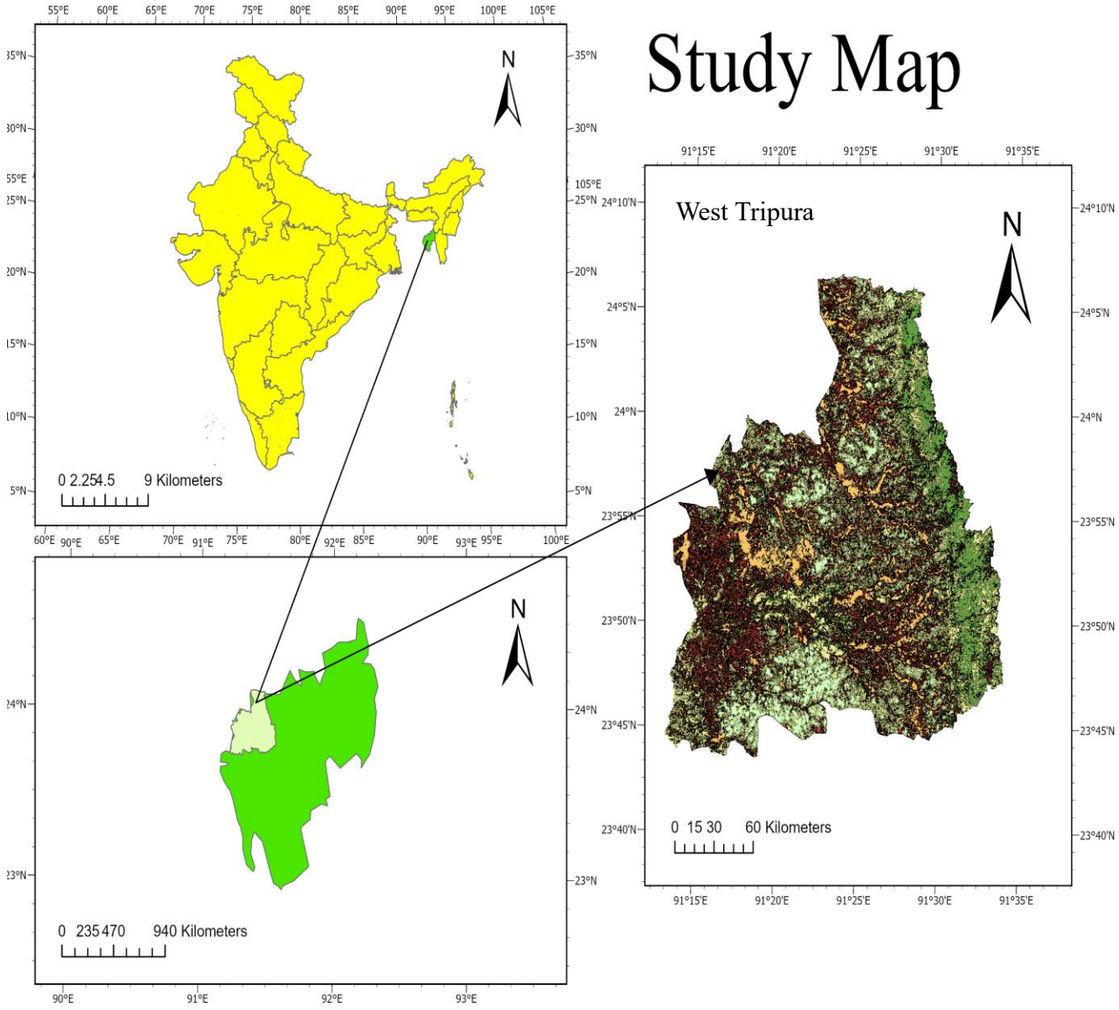
Objective & Study Map

Our Objective

The primary objective of this study is to conduct a comprehensive analysis of land use and land cover (LULC) changes in the West Tripura District from 1990 to 2025. This temporal analysis will provide critical data to inform and guide sustainable urban planning initiatives, ensuring that future development is both economically viable and environmentally responsible.

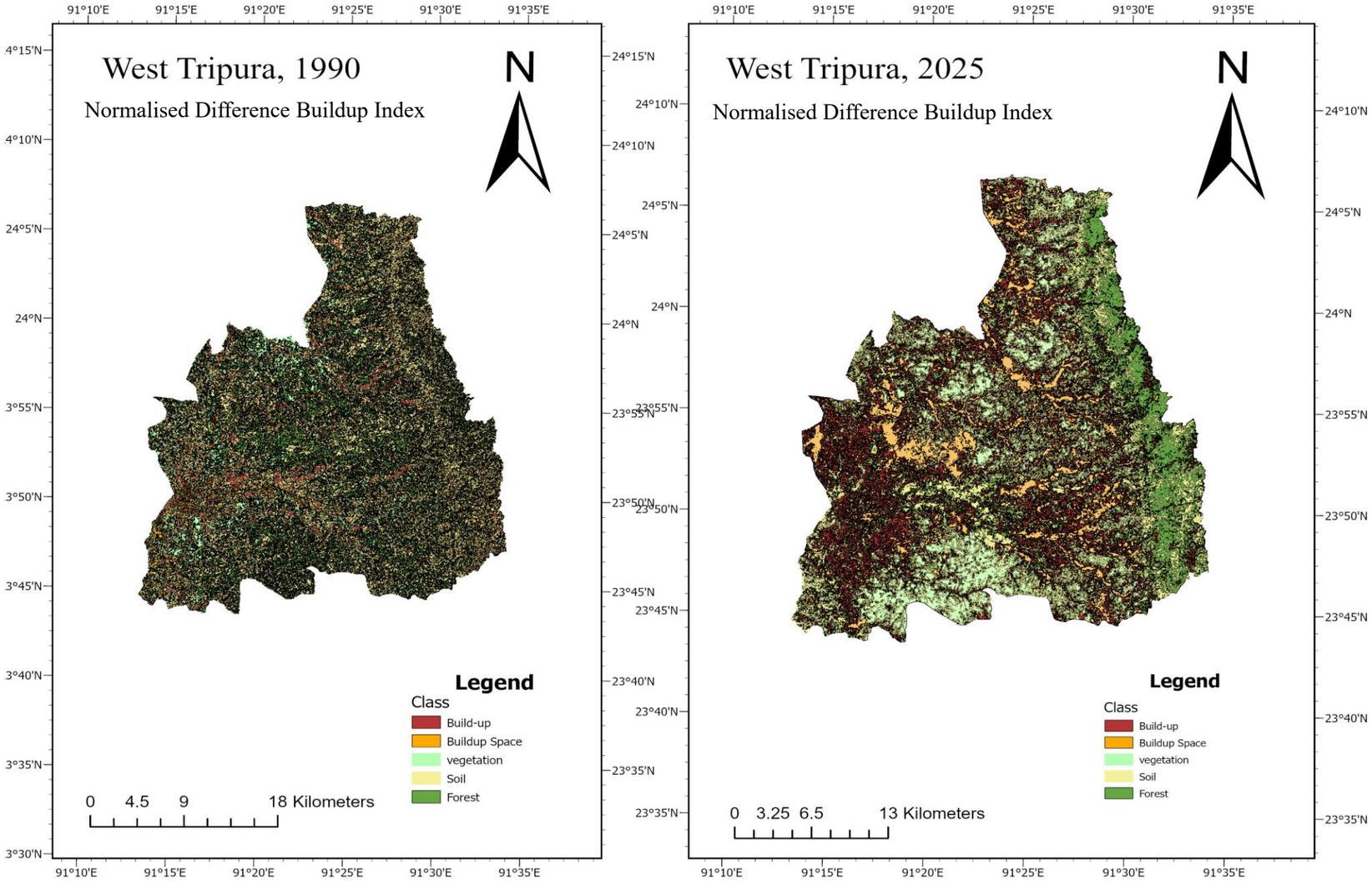
The Study Area: West Tripura District

- Located in the state of Tripura, Northeast India.
- Characterized by a unique blend of urbanizing areas and significant natural ecosystems.
- Known for its rich biodiversity, including diverse forest types and extensive vegetation cover, which are now under threat from urban expansion.
- The district serves as a microcosm for rapid urbanization challenges faced by many regions in developing countries.
- The capital city, Agartala, falls within this district, driving much of the urban growth and developmental activities.



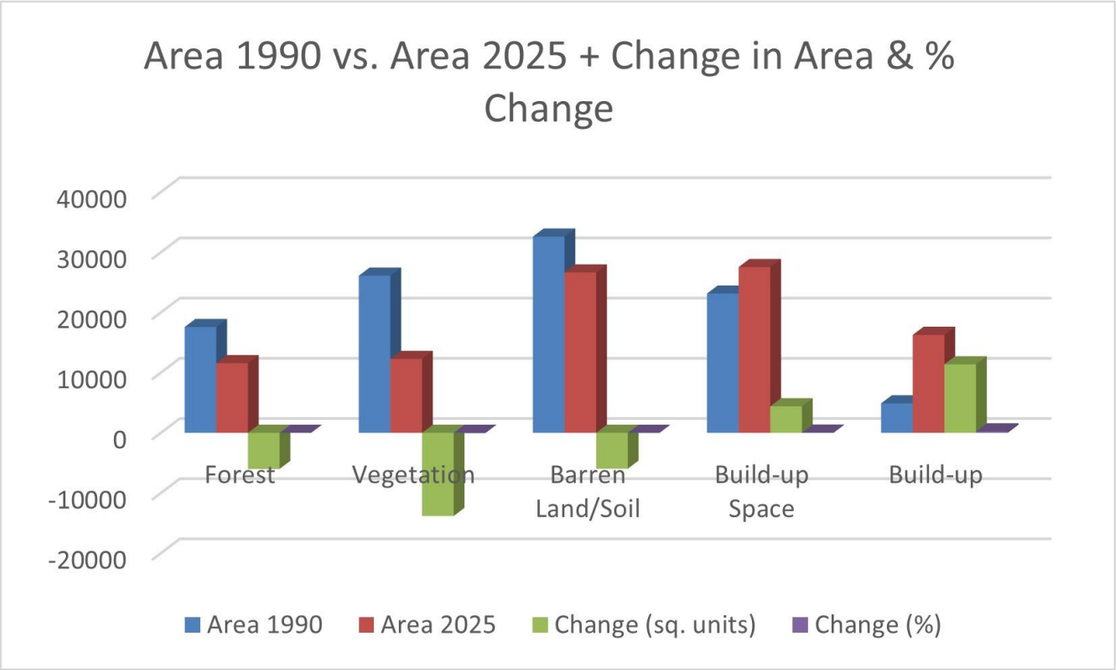
Study Area: Maps & Charts of Transformation

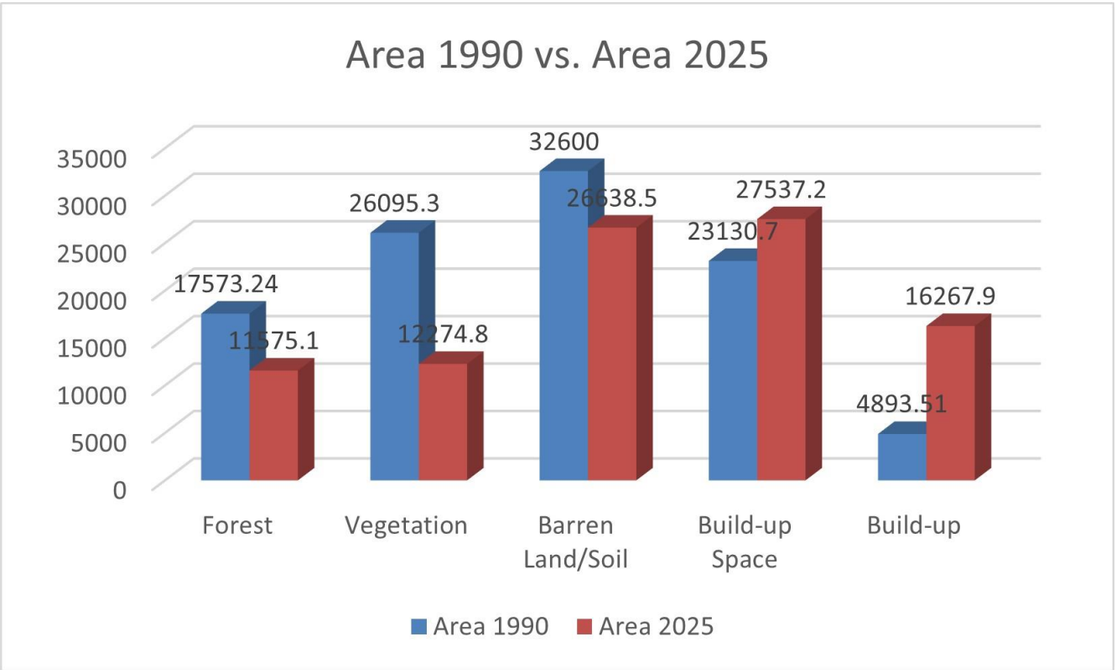
Using ArcGIS Pro, we created clear visualizations that vividly illustrate 35 years of land cover change in West Tripura, turning complex data into an accessible narrative of its transforming landscape.



1. Urban Growth & Land Cover Change: Key Results

Our analysis revealed profound transformations in the land cover of West Tripura District between 1990 and 2025, primarily driven by rapid urban expansion. These results highlight the ecological consequences of unchecked development.





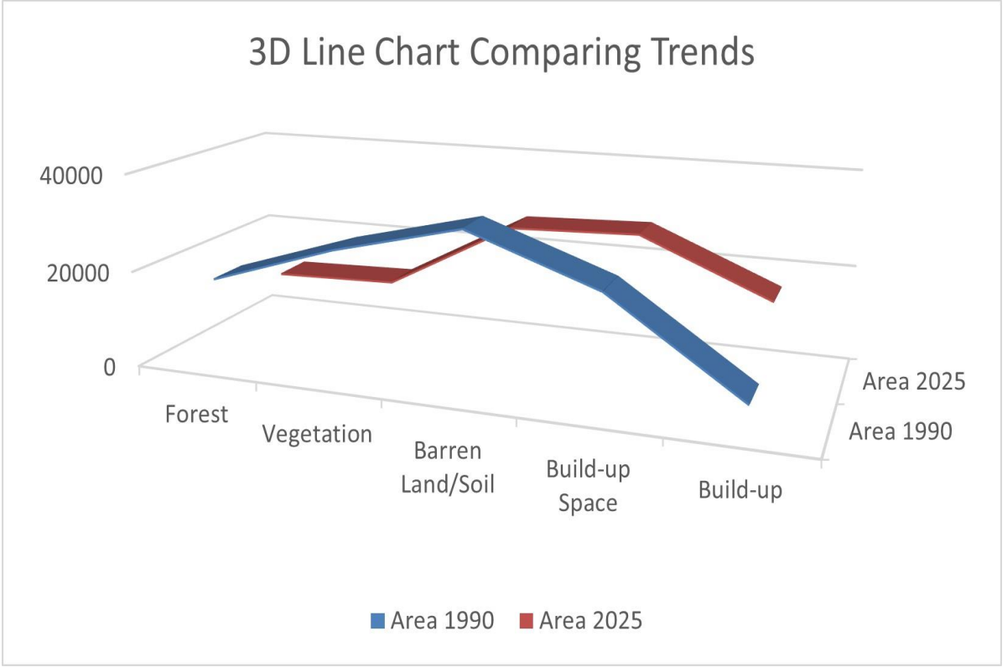
1990 Area (%) 2025 Area (%)

- **Forest Cover Decline:** Forest area fell sharply by **34.13%**, decreasing from **17,573.24 sq. units** in 1990 to **11,575.10 sq. units** in 2025. This substantial reduction highlights the mounting pressure on natural habitats..
- **Built-up Area Expansion:** Built-up areas saw an unprecedented surge of **232.32%**, growing from **4,893.51 sq. units** in 1990 to **16,267.90 sq. units** in 2025. This dramatic rise signals rapid urban sprawl, often encroaching upon formerly forested or vegetated lands.
- **Vegetation Shifts:** General vegetation cover (excluding dense forest) declined by **52.96%**, dropping from **26,095.30 sq. units** to **12,274.80 sq. units**, indicating large-scale conversion to urban or other land uses.
- **Trends Revealed:** The data clearly show a stark inverse relationship—**as urban areas expand, natural cover diminishes**—posing significant challenges to ecological balance and environmental sustainability in West Tripura.

2. City Growth and Land Use Change: Key Results

Our analysis revealed profound transformations in the land cover of West Tripura District between 1990 and 2025, primarily driven by rapid urban expansion. These results highlight the ecological consequences of unchecked development.

Class	Area 1990	Area 2025	Change (sq. units)	Change (%)
Forest	17,573.24	11,575.10	▼ **-5,998.14**	**-34.13%**
Vegetation	26,095.30	12,274.80	▼ **-13,820.50**	**-52.96%**
Barren Land/Soil	32,600.00	26,638.50	▼ **-5,961.50**	**-18.28%**
Build-up Space	23,130.70	27,537.20	▲ **4,406.50**	**+19.06%**
Build-up	4,893.51	16,267.90	▲ **11,374.39**	**+232.32%**



Urban expansion in West Tripura (1990–2025) shows major forest and vegetation loss, offset by rapid build-up growth.

Forests: Declined by ~6,000 sq. units (-34%) – significant habitat loss.
Vegetation: Sharp drop of -52.96%, indicating reduced green cover.

Barren Land/Soil: Slight decrease (-18.28%) – likely due to conversion into built-up areas.
Build-up Space: Moderate increase (+19%) – steady urban growth.

Build-up Areas: Massive surge (+232%) – rapid urbanization & infrastructure expansion.

Discussion: Implications of Normalized Difference Built-up Index

> The observed land cover changes in West Tripura District carry significant implications for its ecological health, socio-economic development, and the future of urban planning.

Ecological Impact

- Loss of vital forest ecosystems leads to reduced biodiversity and habitat fragmentation.
- Diminished green cover impacts local climate regulation and increases urban heat island effects.
- Increased impervious surfaces contribute to higher surface runoff and increased flood risk.

Urban Planning Challenges

- Balancing the imperative for economic growth with the need for environmental conservation remains a critical challenge.
- Informal settlements and uncontrolled sprawl complicate infrastructure development and service provision.
- The need for integrated planning that considers both urban and rural linkages is paramount.

Regional Context

- Similar trends of rapid urbanization and forest conversion are prevalent across other parts of Northeast India.
- This study's findings resonate with broader regional patterns, highlighting a need for concerted regional planning efforts.
- Lessons learned from West Tripura can inform strategies in other rapidly urbanizing areas.

Role of GIS

- GIS, particularly ArcGIS Pro, proves indispensable for data-driven policy formulation and planning decisions.
- It enables precise identification of sensitive areas, monitoring of compliance, and modeling future growth scenarios.
- Empowers planners with the tools to visualize impacts and make evidence-based choices for sustainable development.

Conclusion & Recommendations

The analysis of **West Tripura District (1990–2025)**, conducted using **Landsat satellite imagery** and processed in **ArcGIS Pro**, reveals substantial land cover transformation - marked by **severe forest and vegetation loss** alongside **rapid urban expansion**.

During classification, **water bodies were not detected** in the study area. The dataset was refined using **five spectral bands**, followed by dissolving and reclassification to obtain accurate land cover categories.

Key Takeaways

- **Dramatic Forest Decline:** Forest cover decreased by **34.13%** (17,573.24 → 11,575.10 sq. units).
- **Vegetation Loss:** General vegetation dropped by **52.96%**, indicating major land-use conversion.
- **Explosive Urban Growth:** Built-up areas expanded by **232.32%**, showing rapid urban sprawl.
- **Ecological Trade-off:** Urban growth has occurred primarily at the expense of natural land cover.

Recommendations for Sustainable Development

- **Strengthen Land Use Regulations:** Enforce zoning laws to protect forests and green spaces .
- **Promote Green Infrastructure:** Integrate parks, urban forests, and permeable surfaces into urban design .
- **Continuous Monitoring:** Use remote sensing & GIS for ongoing assessment and adaptive planning..
- **Integrated Planning:** Balance ecological, social, and economic priorities for long-term resilience..

Thank You

We appreciate your time and attention to this critical urban planning case study. We hope this presentation has shed light on the challenges and the potential for geospatial technologies to foster sustainable development.